# Project #7 Instructions
# 11/1/26
# Chris Grimm

**Assignment Instructions**

**Week 7 Project, Part 1: Drawing with Canvas**

**Before You Begin**

**Complete this week's textbook reading and review the related materials before starting.**

**This week, you will complete two hands-on projects worth 50 points each. These instructions are for Part 1.**

**1. Add a Canvas to Your Home Page**

**Add a <canvas> element to your homework home page. Use JavaScript to draw at least six separate items, including:**

- **One arc or circle**
- **One triangle**
- **One rectangle**
- **At least two lines**
- **At least one additional shape or line of your choice**

**Give each item a defined stroke, fill, or both, as appropriate. Lines should have a visible stroke.**

**You may arrange the items into a picture or an abstract design. Make each required item easy to identify.**

**2. Apply a Composite Operation**

**Make at least two shapes overlap and apply a composite operation that visibly affects how they appear together.**

**Refer to pages 262–263. Experiment with the drawing order and the operation you select so you understand how they affect the result.**

**3. Use a Transformation**

**Use at least one transformation when drawing a shape. You may translate, rotate, or scale the drawing coordinates.**

**Refer to pages 256–260. Remember that a transformation changes the drawing coordinate system; it does not move the <canvas> element itself on the page.**

**4. Validate and Test Your Work**

**Validate your HTML and correct any errors. Also check your browser's developer console for JavaScript errors, since HTML validation does not verify that your drawing code works.**

**Upload your revised HTML, JavaScript, and any supporting files to the student server. Open your published page and confirm that:**

- **All six required drawing items appear.**
- **The overlapping shapes demonstrate your composite operation.**
- **Your transformation produces the intended result.**
- **Your existing page content and navigation still work.**

**Submit Your Project**

**Submit the following through this assignment in Blackboard:**

1. **The URL to your updated homework home page on the student server.**
2. **The transformation you used, identifying the shape it affects.**
3. **The composite operation you used, identifying the overlapping shapes and describing its visible effect.**

**Take time to experiment and make the drawing your own. If you get stuck, please reach out. I am happy to help!**

**Sources:**

index.html

script.js

style.css

**Live URL:** http://157.245.246.41:8014/index.html

**Transformation Used:**

I used a translation and rotation transformation. It affects the purple square in the top right, rotating it into a diamond.

**Composite Operation Used:**

I used the “xor” composite operation on an orange circle and an overlapping cyan square. Where the square overlaps the circle, it cuts out the color and turns transparent, leaving a square cutout that show the dark background underneath.

**Front End:**

